DAY 2: DHCP CONFIGURATION

DHCP Process and Cisco Packet Tracer Practical

Introduction: What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol. It automatically provides network settings such as an IP address, subnet mask, default gateway, and DNS server to a client device. This reduces manual configuration and helps prevent address conflicts.

The four main DHCP steps are called DORA:

1. Discover: The client broadcasts a message to find a DHCP server.
2. Offer: The DHCP server offers an available IP address and network settings.
3. Request: The client requests the offered address.
4. Acknowledge: The server confirms the lease and sends the final settings.

Task 1: Configure One DHCP Server and Two PCs

Aim: To configure a DHCP server in Cisco Packet Tracer and confirm that two PCs receive different IP addresses automatically.

Example topology: Server0 and PC0 and PC1 connected to one switch.

Suggested addressing plan:

Device / setting	Example value
Network	192.168.1.0 / 24
Server0 static IPv4	192.168.1.2
Subnet mask	255.255.255.0
Default gateway (if router exists)	192.168.1.1
DHCP pool start address	192.168.1.100
Maximum clients	At least 2

Steps:

5. Place one Server, one Switch, and two PCs in the Packet Tracer workspace. Connect each device to the switch using Copper Straight-Through cables.
6. On Server0, open Desktop > IP Configuration and assign static IP 192.168.1.2 with subnet mask 255.255.255.0.
7. Open Server0 > Services > DHCP. Turn DHCP On, set a pool name, default gateway if used, DNS server if needed, start IP 192.168.1.100, subnet mask 255.255.255.0, and maximum users to at least 2. Click Add or Save as shown in your version.
8. On PC0, open Desktop > IP Configuration and select DHCP. Repeat for PC1.
9. Check that each PC receives a valid address and that the two addresses are different. If needed, wait briefly or click DHCP again.

Task 2: DHCP DORA Sequence Diagram

The following sequence diagram shows the message direction between a DHCP client and a DHCP server.

Step	Sender → Receiver	Message / purpose

1. Discover	Client → DHCP Server(s)	Client broadcasts to find a DHCP server.
2. Offer	DHCP Server → Client	Server offers an available IP address and settings.
3. Request	Client → DHCP Server(s)	Client requests the offered IP address.
4. Acknowledge	DHCP Server → Client	Server confirms the address lease and configuration.

Simple arrow diagram:

DHCP CLIENT **DHCP SERVER**

Discover ---------->

Offer <----------

Request ---------->

Acknowledge <----------

Task 3: Restrict the DHCP Pool to One IP Address

Aim: To observe what happens when two PCs request an address but the DHCP pool contains only one available address.

Steps:

10. Use the Task 1 topology with two PCs connected to the switch and the DHCP server.
11. Open Server0 > Services > DHCP and change the pool so only one usable address is available. For example, set the start IP to 192.168.1.100 and maximum users to 1. Ensure the address is not already leased to another device.
12. Renew the DHCP setting on both PCs by selecting Static and then DHCP, or use the relevant renew option available in Packet Tracer.
13. Check the IP configuration of both PCs. One PC should receive the only available address; the other should fail to obtain a valid lease or may show an automatic/private fallback address, depending on the simulation.

Explanation

The DHCP pool has only one available IP address, so the server can lease that address to only one client at a time. The second PC cannot receive an address from this pool until another address becomes available or the pool is expanded.

Screenshot: [Paste the DHCP pool setting and both PC results here]

Task 4: Zomato-Style Restaurant Wi-Fi Scenario

Scenario: A restaurant provides Wi-Fi for customers. A laptop and a generic PC representing a smartphone connect to the same access network and receive IP addresses automatically from a DHCP server.

Suggested Packet Tracer topology: DHCP Server + Switch + Laptop + PC (representing smartphone). If using a wireless router/access point in your design, connect the client devices to that network as supported by your Packet Tracer device choices.

Steps:

14. Place a DHCP Server and a Switch in the workspace. Add a Laptop and a PC to represent a smartphone; connect them to the same LAN using the available Packet Tracer connections.
15. Configure the server with a static IP such as 192.168.10.2 and subnet mask 255.255.255.0.
16. In Services > DHCP, turn the service On. Use network 192.168.10.0/24, start address 192.168.10.100, and allow at least two clients.

17. Set both client devices to obtain an IP address using DHCP. Wait for the leases to be assigned.
18. Record the actual IP addresses and capture a screenshot of the topology and client configurations.

Why DHCP is useful:

DHCP automatically gives each customer device its own IP address and required network settings. Restaurant staff do not need to configure every customer's device manually, and the DHCP server helps manage addresses for many users.

Conclusion

In this practical, I learned how DHCP assigns IP addresses automatically, how the Discover, Offer, Request, and Acknowledge steps work, why a one-address pool can serve only one client at a time, and how DHCP can simplify customer Wi-Fi in a restaurant. I will include actual Packet Tracer screenshots and assigned IP addresses before submission.